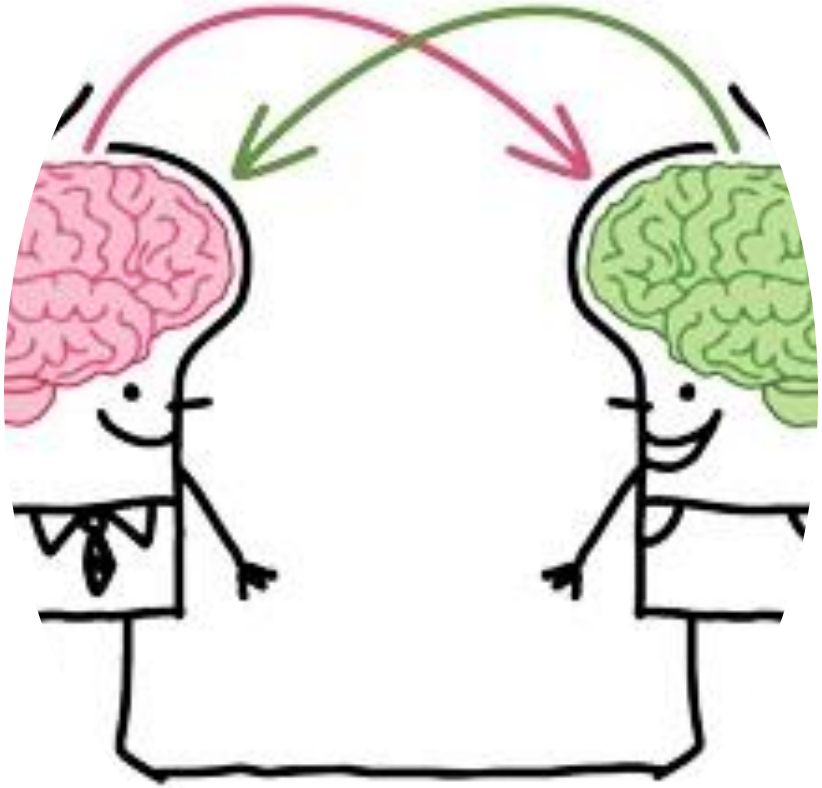
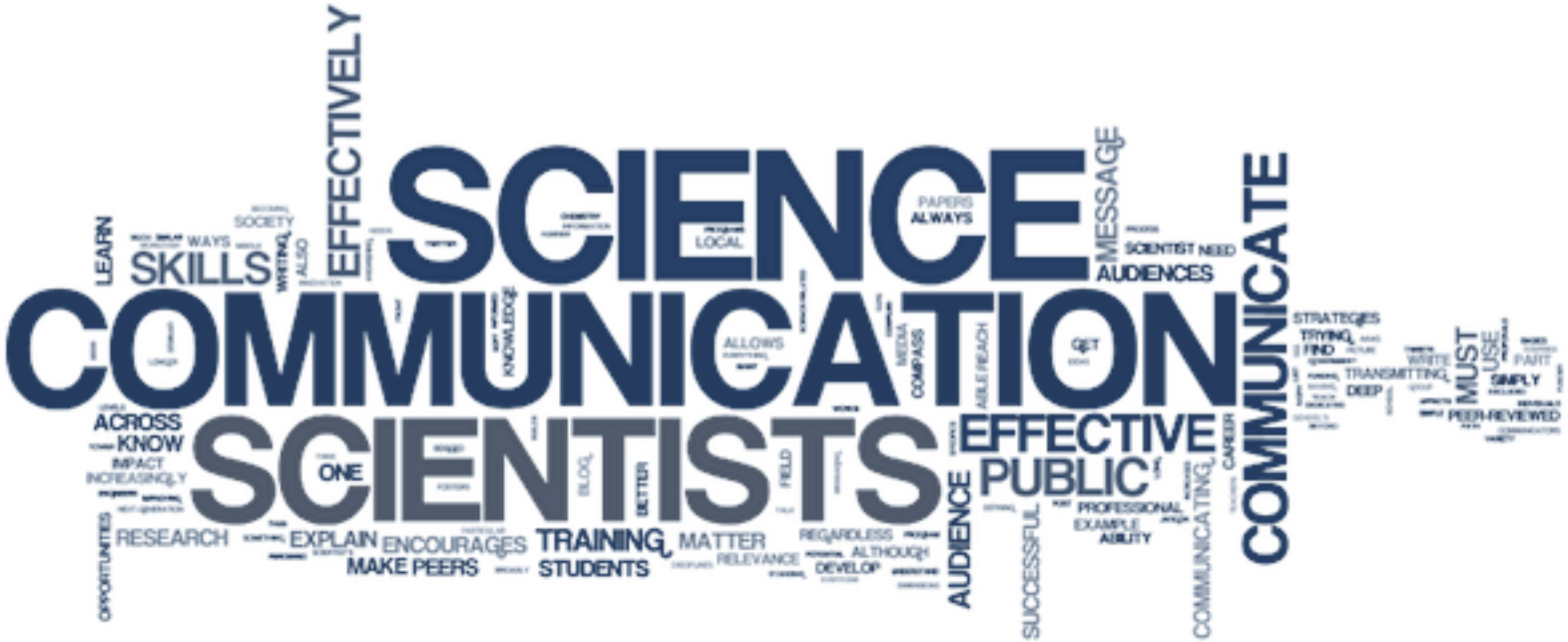
Data Visualization and Plots in R

Carolina Sarmiento, USF
October, 2025



Objective:
Communicate our science



- Papers and proposals
- Talks
- Teach
- Outreach
- Social media + Journalists



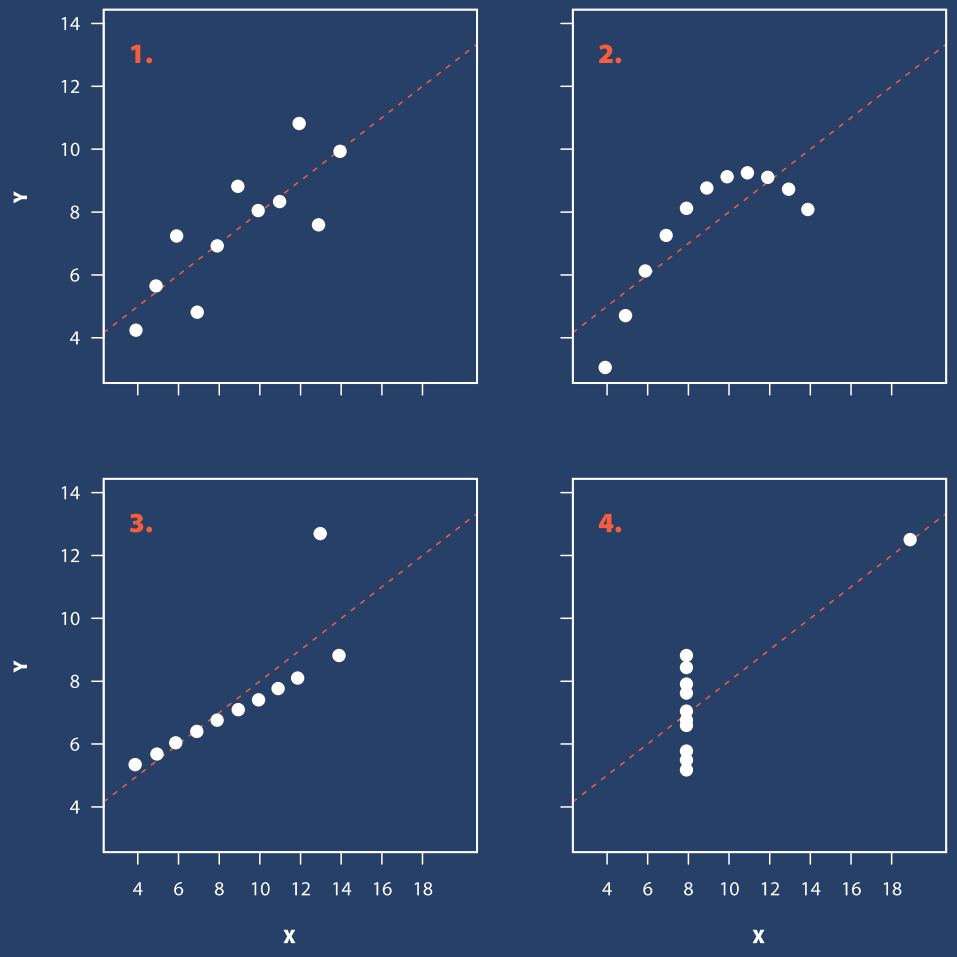
- Papers and proposals
- Talks
- Teach
- Outreach
- Social media + Journalists

" To be a successful scientist, you must be an effective communicator"

Graphics

Not only data, also patterns

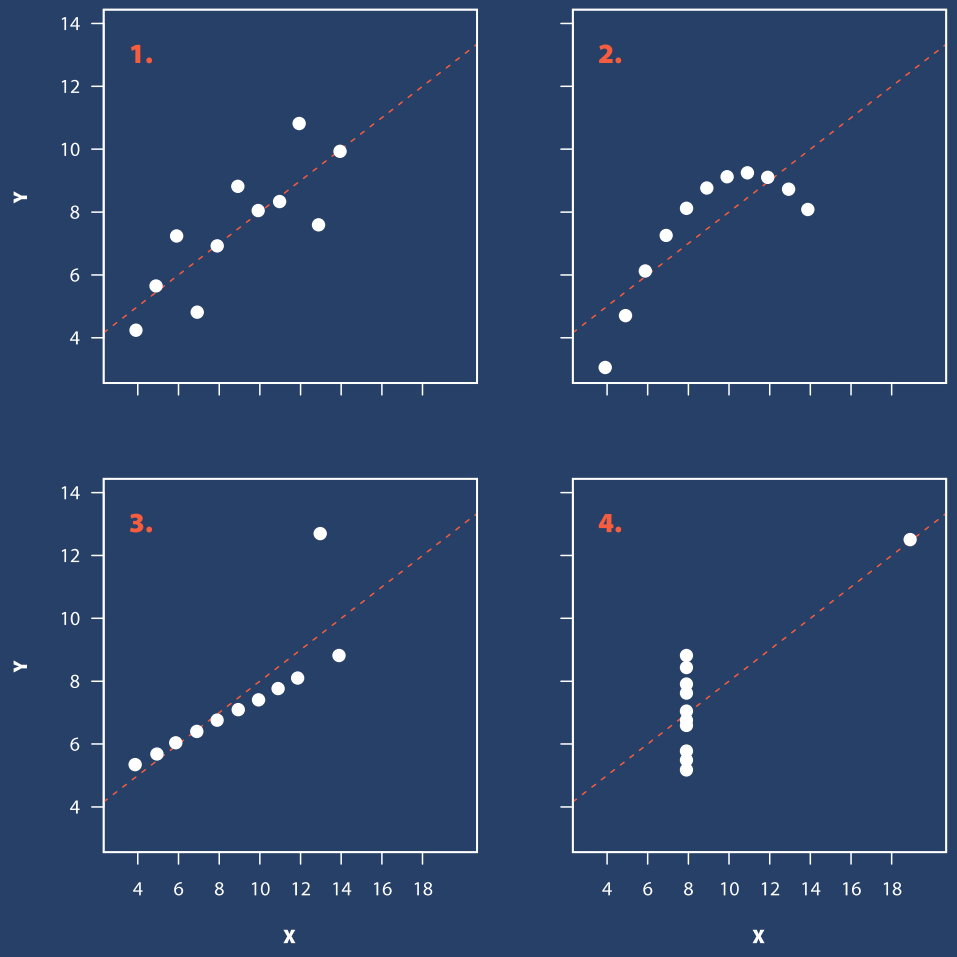
Anscombe's quartet:



Graphics

Not only data, also patterns

Anscombe's quartet:



Same:

- Number of points
- Mean (X, Y)
- SD (X, Y)
- Regression line
- Standard error of the slope
- Sum of squares
- Residual sum of squares
- Correlation coefficient
- r^2

A good plot should:



Show the data

Tell the truth



Encourage relevant comparisons

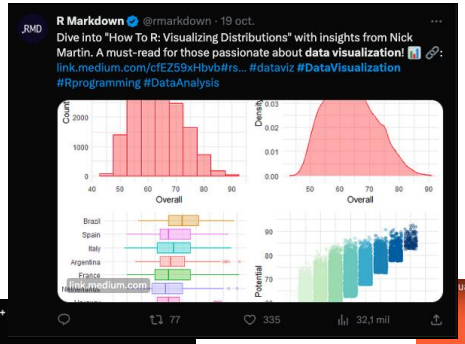


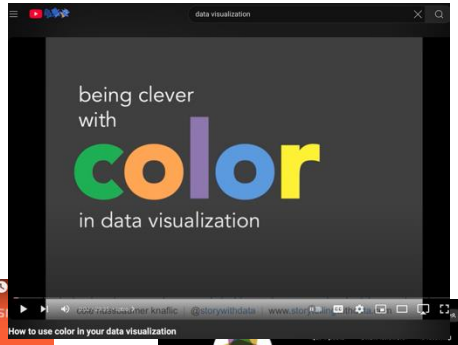
Help the viewer think about the information, not methodology or design

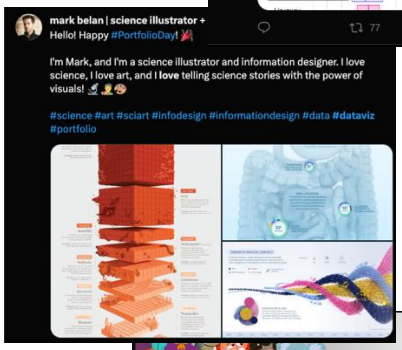


Make large datasets coherent

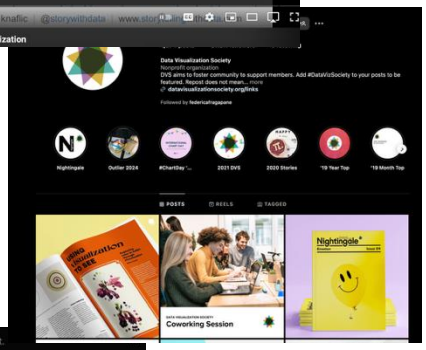
DV is popular

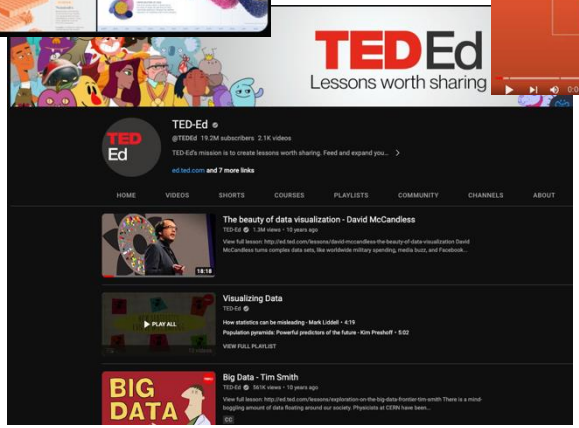


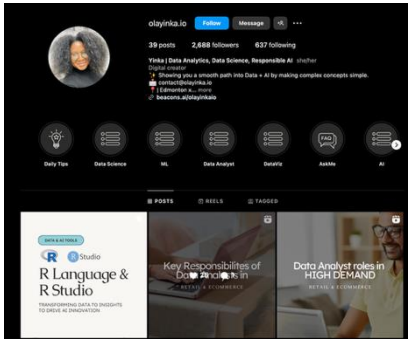


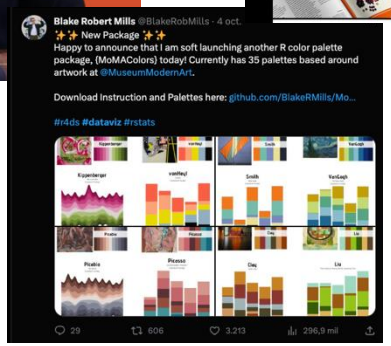












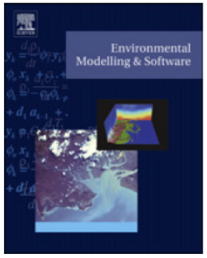
Guidelines for Data Visualization

Environmental Modelling & Software 26 (2011) 822–827

Contents lists available at ScienceDirect

Environmental Modelling & Software

journal homepage: www.elsevier.com/locate/envsoft



Short communication

Ten guidelines for effective data visualization in scientific publications

Christa Kelleher*, Thorsten Wagener

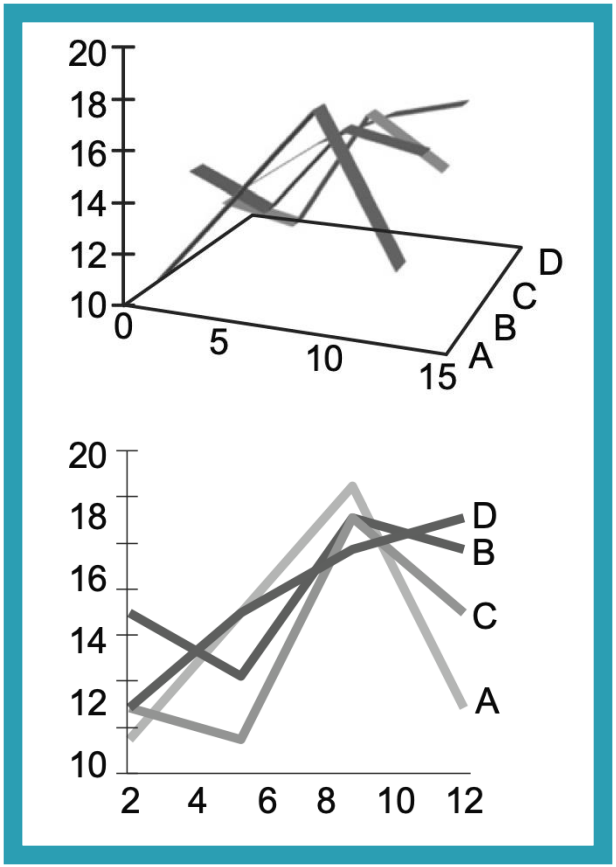
Department of Civil and Environmental Engineering, The Pennsylvania State University, 212 Sackett Building, University Park, PA 16802, USA

* Guideline: General principle that can be applied most of the time, but to which there are exceptions

1. Less is More

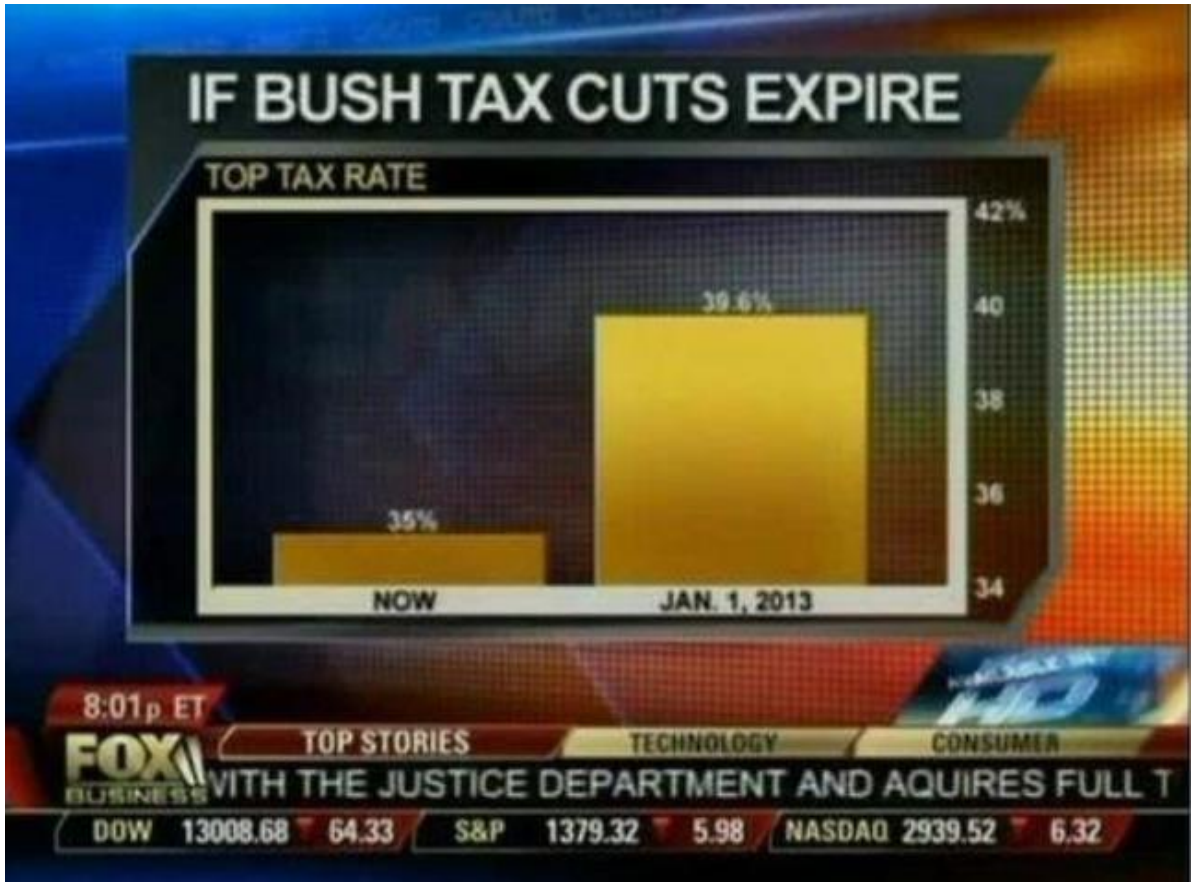
Create the simplest graph that conveys the information you want to convey

- Remove redundancy (e.g. using different colors and types of points to show the same information)
- Make sure the reader can distinguish among properties (size, color, line thickness)



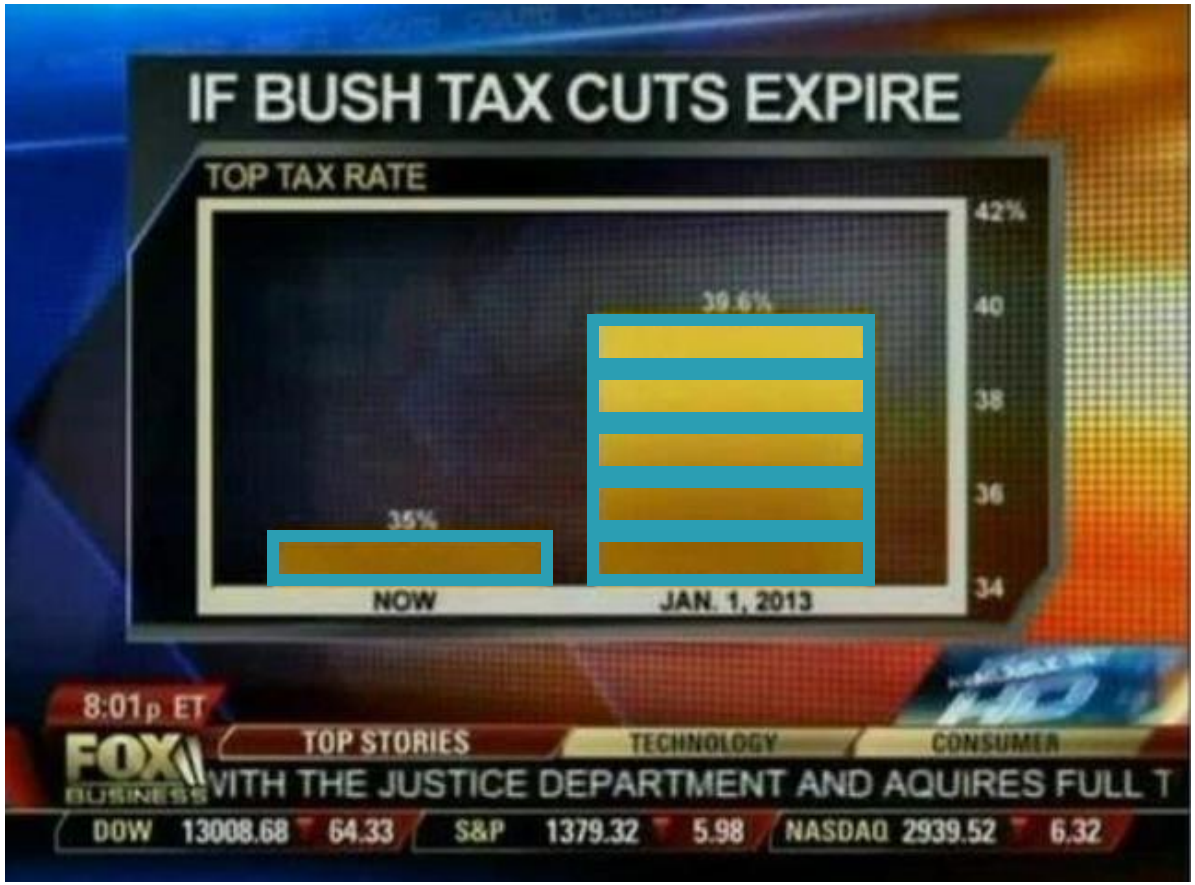
2. Be precise

Select meaningful axis ranges



2. Be precise

Select meaningful axis ranges



After a tax cut expire, there would be a **huge** increase in tax rate

2. Be precise

Select meaningful axis ranges

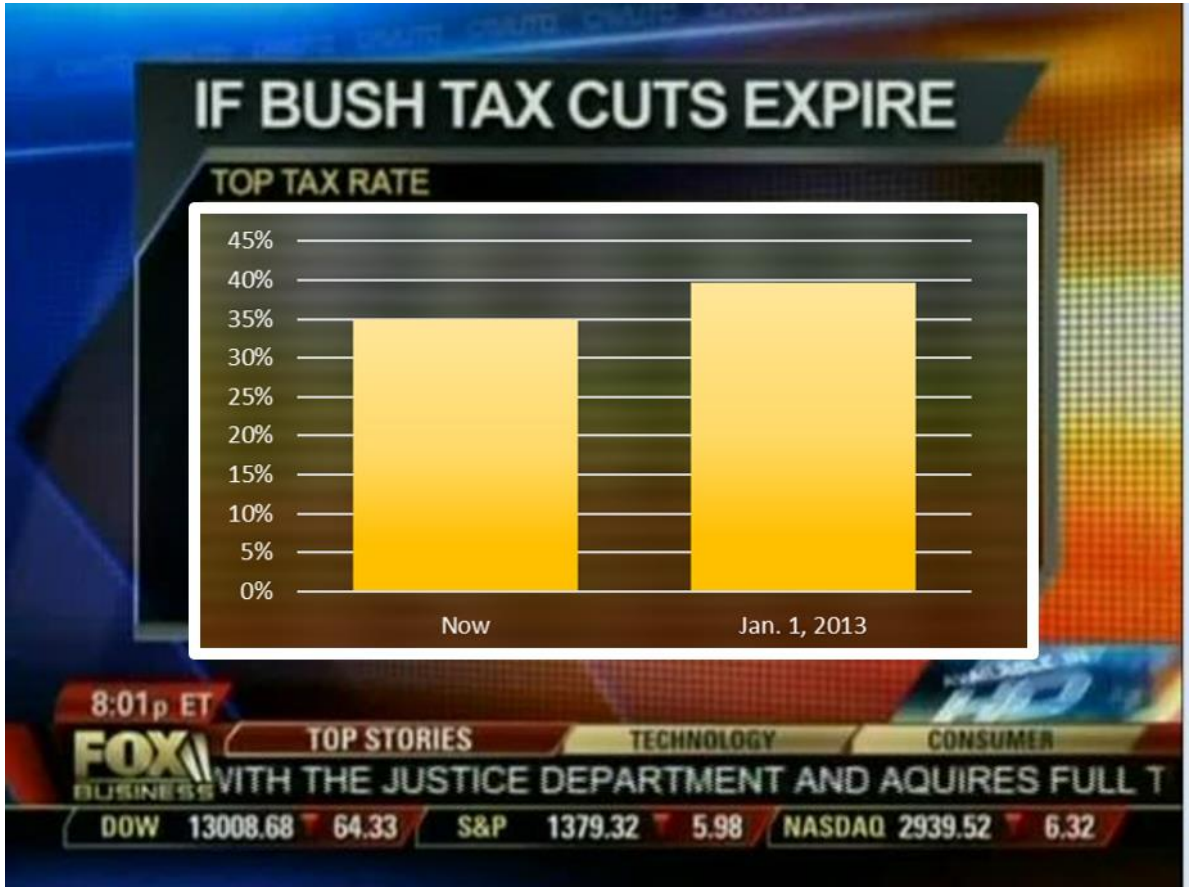


After a tax cut expire, there would be a **huge** increase in tax rate

BUT

2. Be precise

Select meaningful axis ranges



After a tax cut expire, there would be a **huge** increase in tax rate

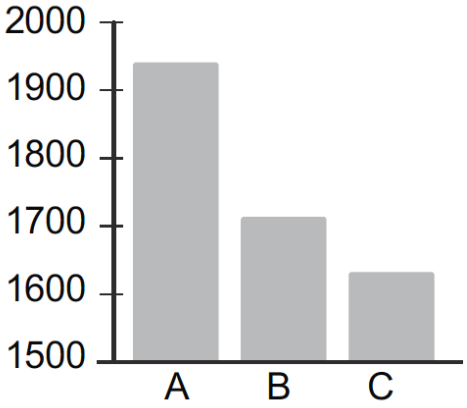
BUT

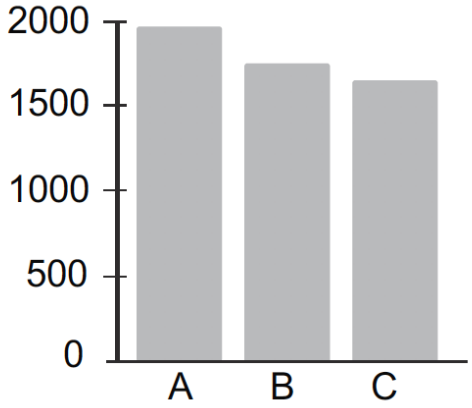
A tax rate of 35% is **not drastically different** from a tax rate of 39.6%

Source: Towards Data Science

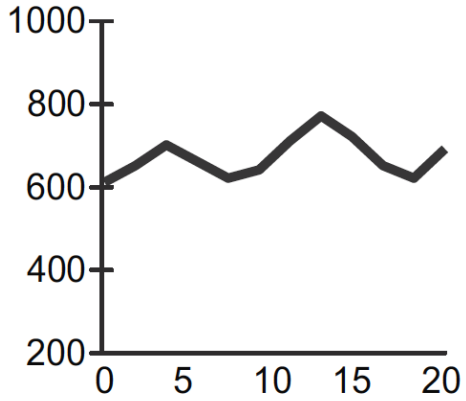
2. Be precise

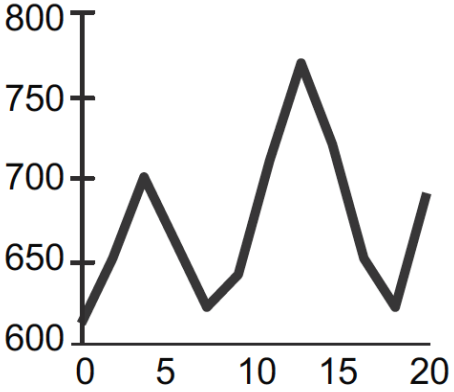
Selecting the range depends on the graph's purpose and type:





Absolute magnitudes are important

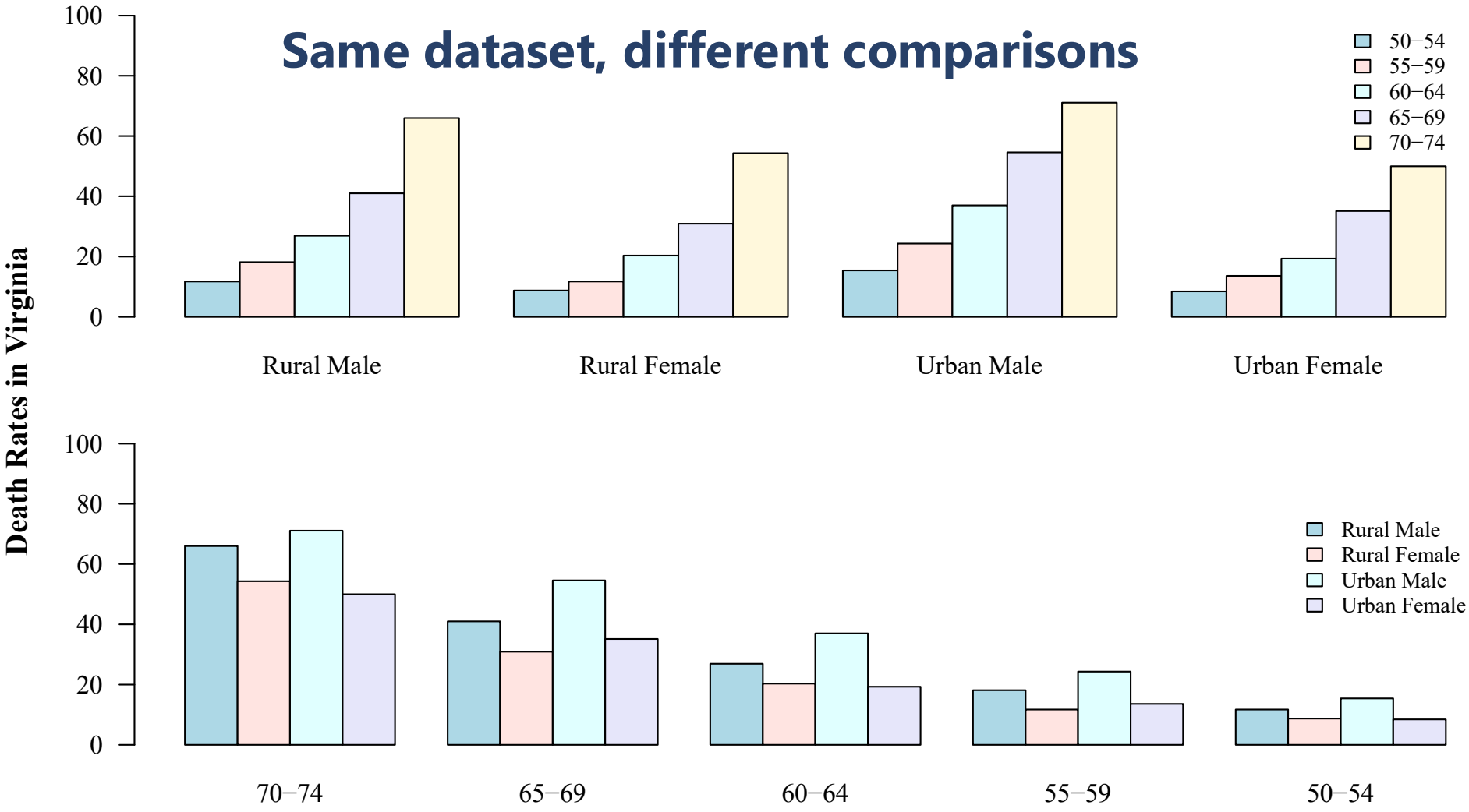




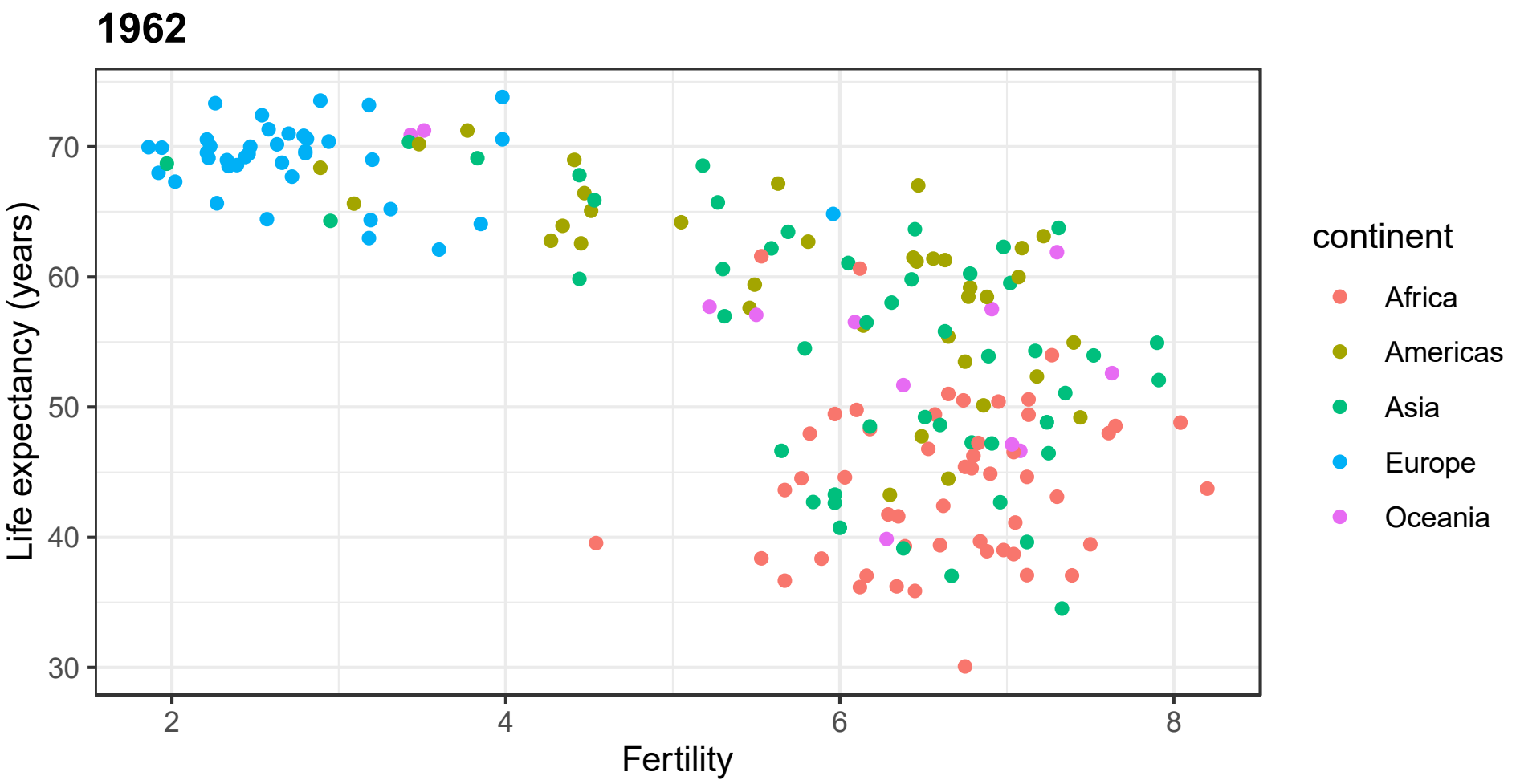
Relative magnitudes are important

Setting the limits of the plot as close as possible to the range of the dataset: improves visualization of variability and avoid white (wasted) space

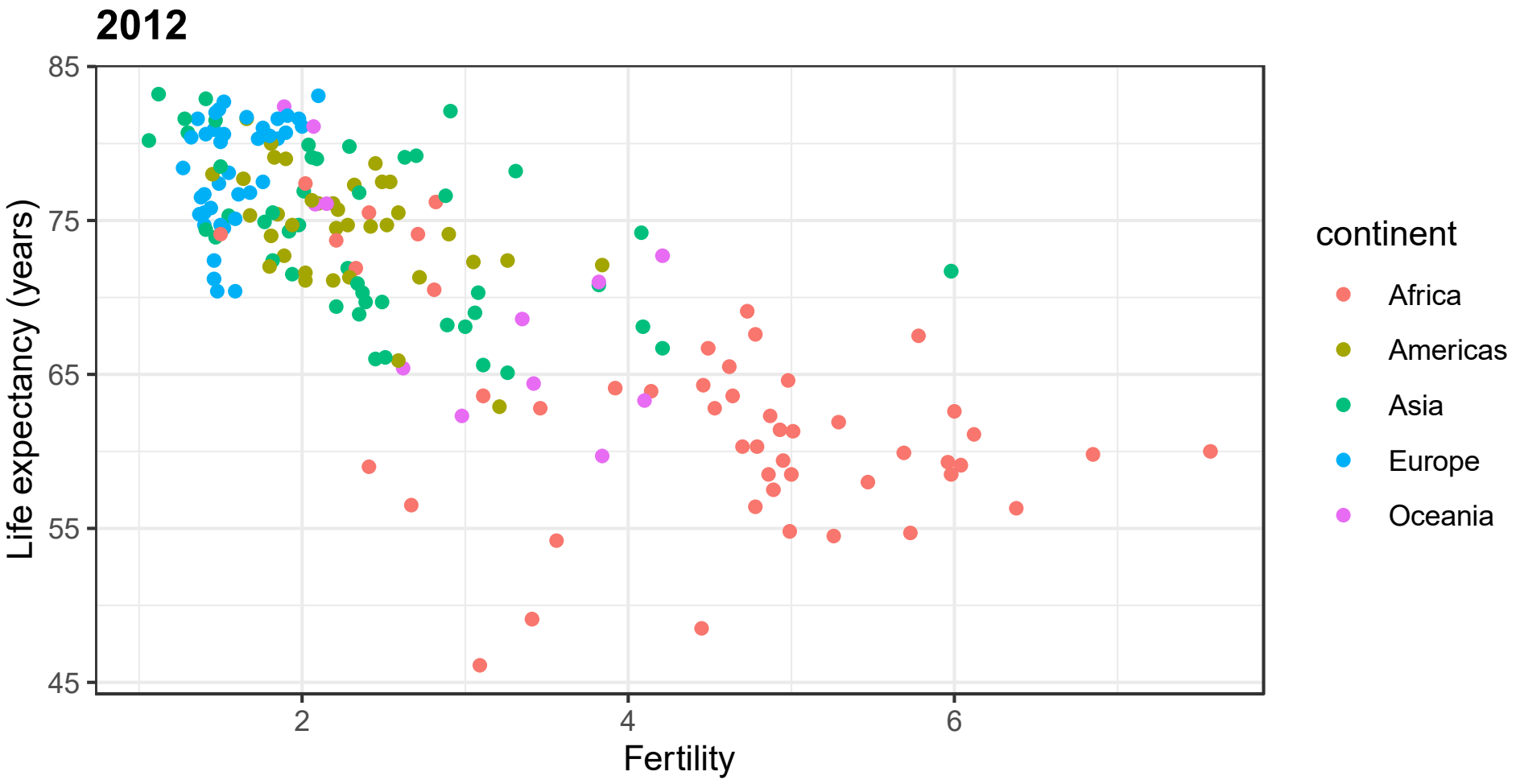
3. Relevant comparisons



3. Relevant comparisons

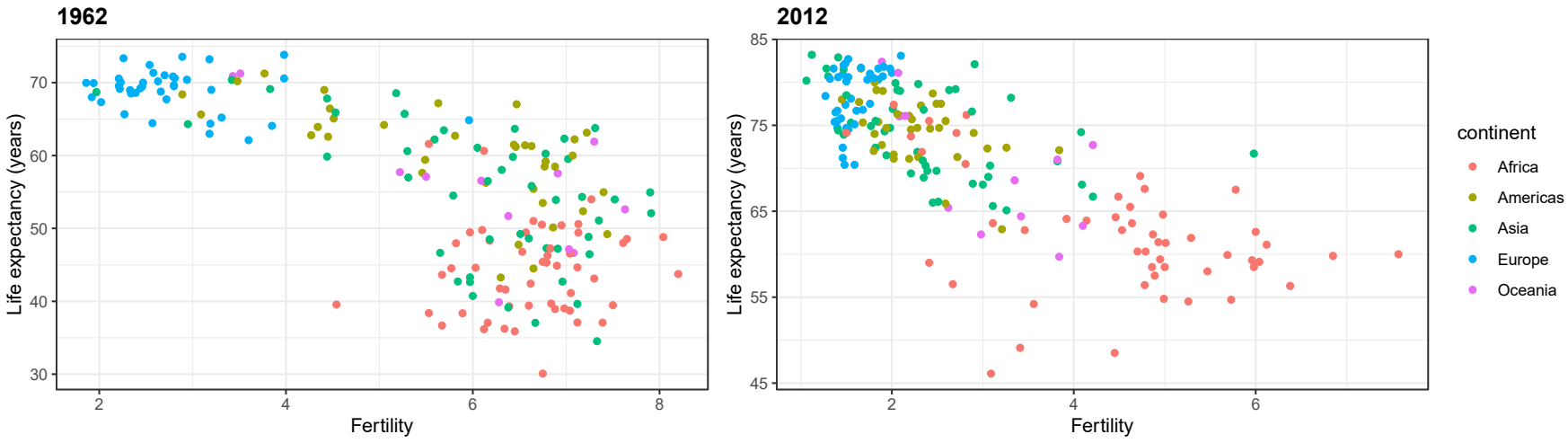


3. Relevant comparisons



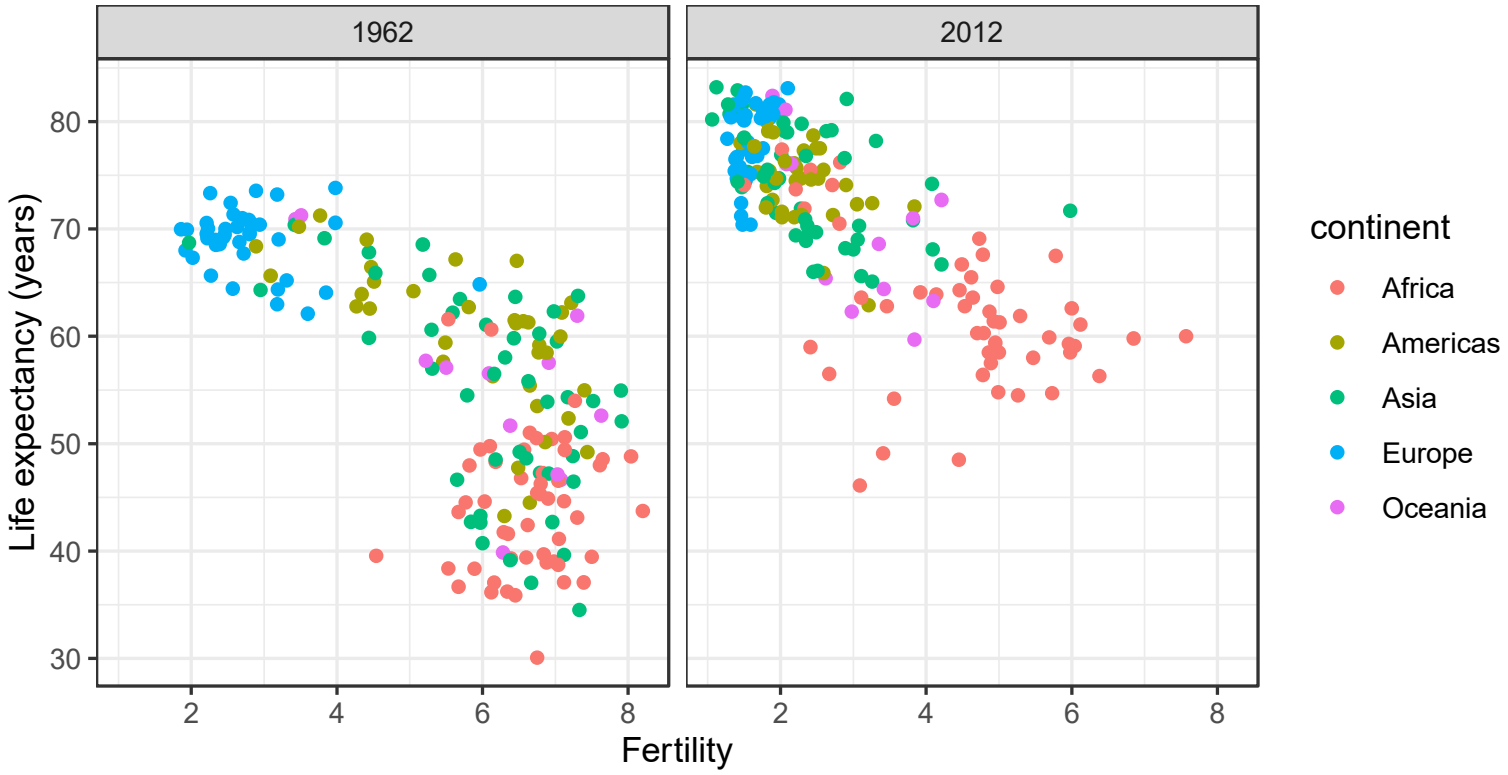
3. Relevant comparisons

It has changed, but not much...



3. Relevant comparisons

Yes, it has changed



4. Be careful

Choosing colors and patterns

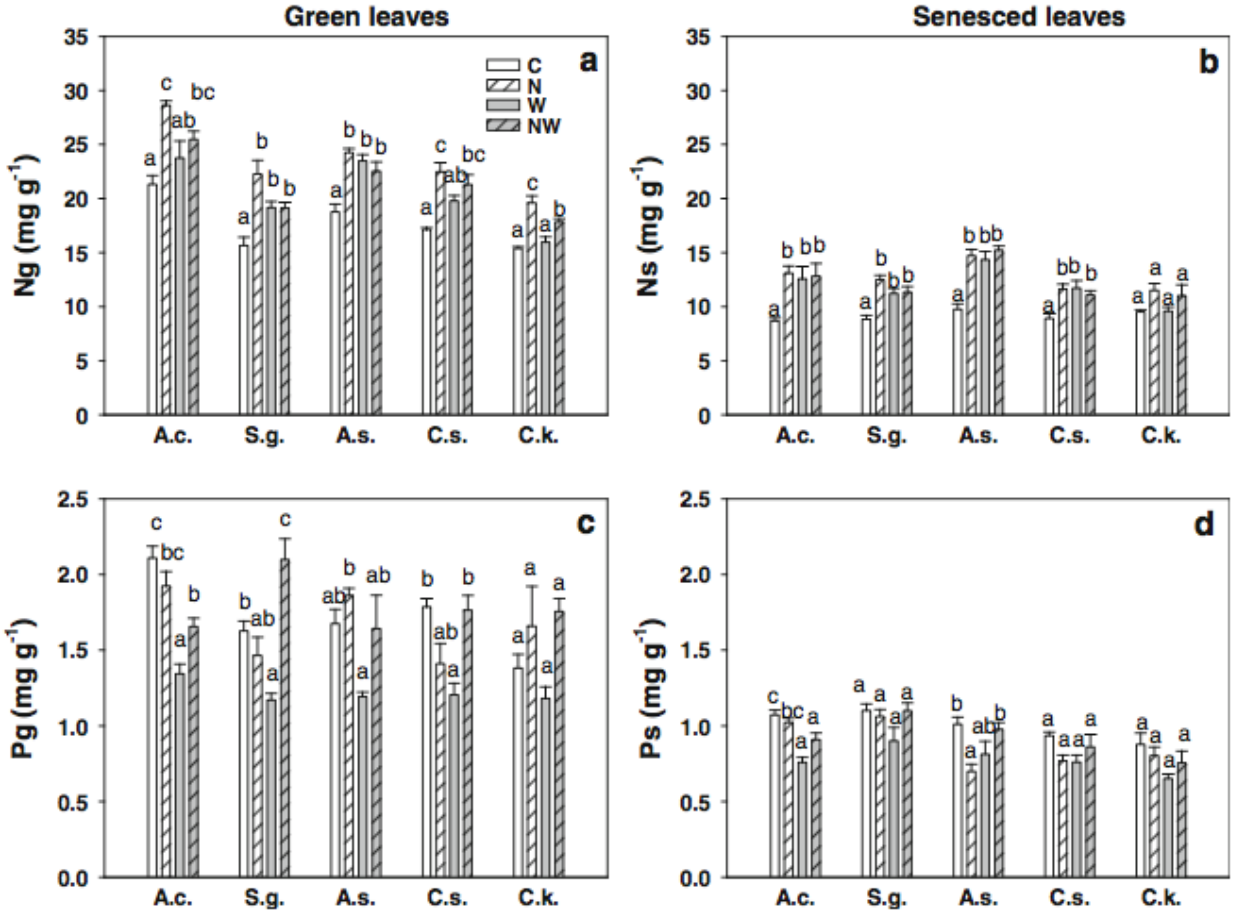


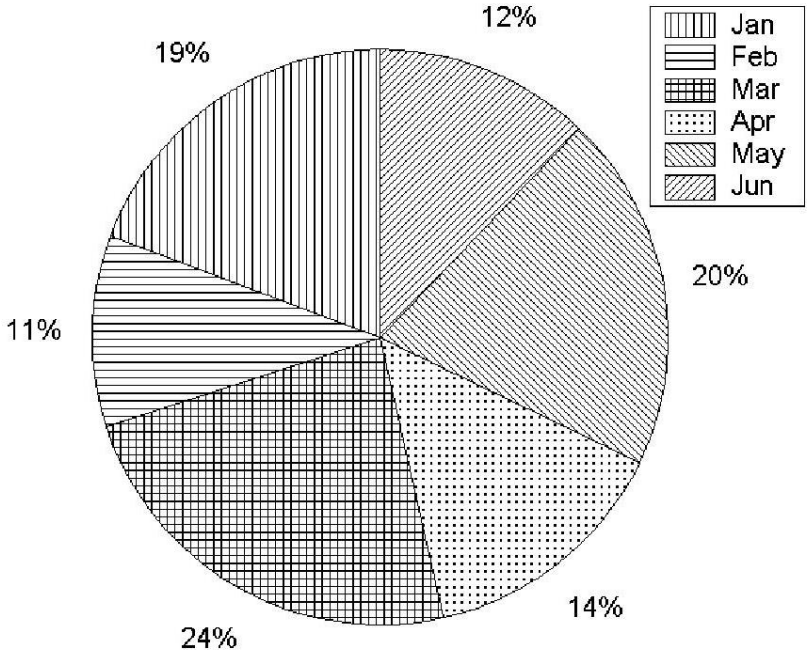
Fig. 2 Responses of N (**a**, **b**) and P (**c**, **d**) concentrations in green (Ng, Pg) and senesced leaves (Ns, Ps) of five plant species to N and water addition. *Error bars* +1SE. *Bars* within species with *dissimilar letters* denote significant difference ($P < 0.05$). *A.c. Agropyron cristatum*, *S.g. Stipa grandis*, *A.s. Achnatherum sibiricum*, *C.s. Cleistogenes squarrosa*, *C.k. Carex korshinskyi*

Oecologia, 2012

4. Be careful

Choosing colors and patterns

Avoid excessive use of grids, patterns, and decorative forms



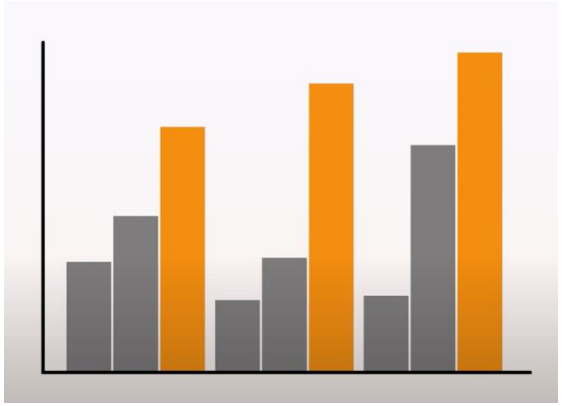
4. Be careful

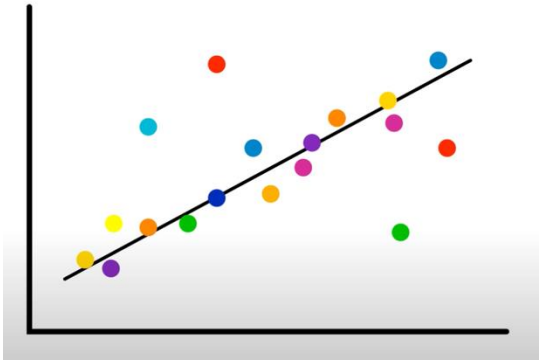
Choosing colors and patterns

Color can enhance and clarify your figures
(if correctly selected)

or

Color can be confusing, hide the meaning of your data or distort it



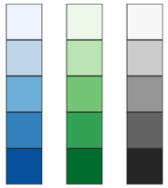


4. Be careful

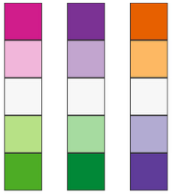
Choosing colors and patterns

Use a color scheme that matches the type of data

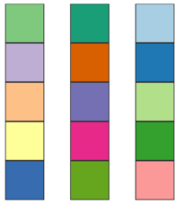
Sequential: used for quantitative data varying from low (lighter) to high (darker). Variation of a unique color.



Diverging: used to highlight contrasts between low and high values relative to a median value.



Qualitative: used mainly for discrete or categorical data. Rapid variation of colors without reference to magnitude.



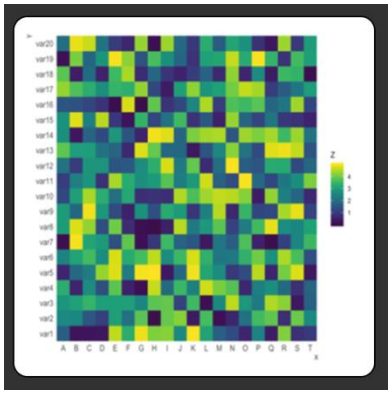
Check ColorBrewer (Website and R package)

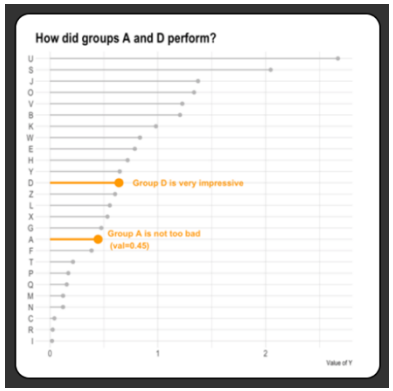
General suggestions

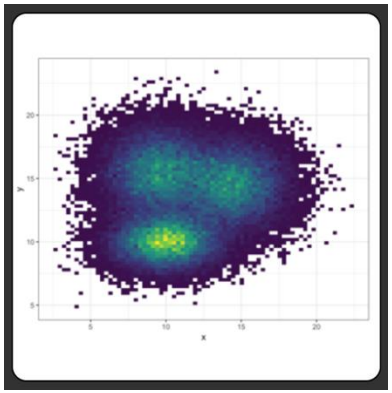
- Use consistent format
- Use readable fonts (size, color, font choice)
- Avoid cluttered legends. Where possible, add labels directly to the elements of the plot
- Make sure the axes are scaled properly
- Axes should be properly labelled, do not forget to specify units
- Plots should be self-explanatory. If needed, include detailed captions that match the graphic
- Choose color wisely (colorblindness = colorblindness)

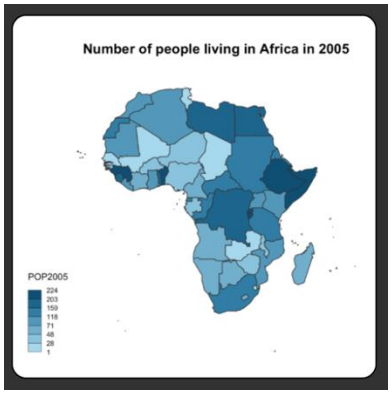
Plots in R

Graphic capabilities: one of the strengths in R

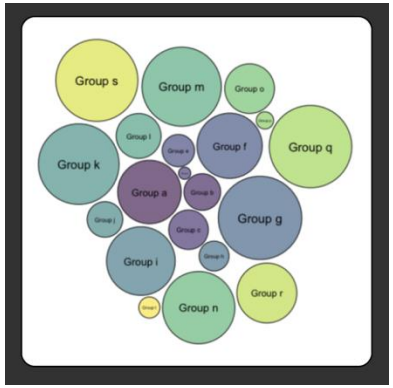


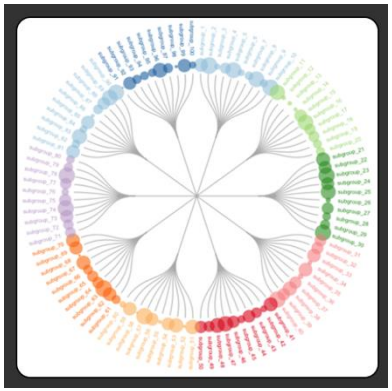


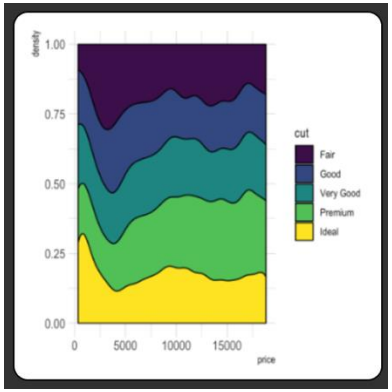












Plots in R (base)

Useful functions

plot( ) # create a new plot of any type
hist( ) # create a histogram
lines( ) # add straight lines to an existing plot
points( ) # add points to an existing plot
text( ) # add text to an existing plot
abline( ) # add a line of the form ax + b to an existing plot
segments( ), **arrows( )**, **rect( )**, **polygon( )**
add line segments, arrows, or rectangles to an existing pl
legend( ) # add a legend to an existing plot

lines 1 - 42

Plots in R (base)

? par: Set graphical parameters

[also arguments inside plot ( ), line ( ), etc.]

pch = point types
main = main title
col = set colors
xlim = set limits on the x-axis
ylim = set limits on the y-axis
xlab = set title (label) for the x-axis
ylab = set title (label) for the y-axis
las = style of axis labels (las = 1, horizontal axes labels)
lty = line type (0=blank, 1=solid (default), 2=dashed, 3=dotted, 4=dotdash, 5=longdash, 6=twodash)
mai = margin size in inches
font.lab = font to be used for x and y labels (1= plain text (default), 2=bold, 3=italic, 4=bold italic)

Plots in R (base)

Let's reproduce the Anscombe's quartet plot

Use the function *pdf ( )*:

- Starts the graphics device driver for producing PDF graphics (also *svg( )*, *postscript( )*)
- Once you start it, you won't be able to see your plot on RStudio plot window
- Use ***dev.off ( )*** to close the current device. This will save your plot in pdf format, inside your WD

lines 44 - 131

Plots in R (ggplot2)

- "**ggplot2** is a plotting package that makes it simple to create complex plots from data in a data frame"
- "The **ggplot2** package offers a powerful graphics language for creating elegant and complex plots. Its popularity in the R community has exploded in recent years"
- "Mastering the **ggplot2** language can be challenging"
- "**ggplot2** is the most elegant and aesthetically pleasing graphics framework available in R"

Plots in R (ggplot2)

It is very different from base graphics

Basic Structure:

1. The setup
2. The layers
3. The labels
4. The theme
5. The facets

Plots in R (ggplot2)

1. The setup: tell ggplot what dataset to use and set x and y variables

   Use the functions

   ggplot ( ): Create a new ggplot

   and

   aes( ): Set the aesthetics mappings

   "How variables are mapped to the visual properties of the geoms"

```
p <- ggplot(iris, aes(x=Sepal.Width, y=Sepal.Length))
```

Plots in R (ggplot2)

2. Add the layers (also "geoms" or geometric objects)

using the + symbol

+ geom_point( )

There are many **geom_** functions: geom_boxplot, geom_abline, geom_histogram, geom_errorbar, etc. Check the *ggplot* Reference page

lines 149 - 151

Plots in R (ggplot2)

3. **Add some labels:** you might want to add the plot's main title and perhaps change the X and Y axes titles.

   This can be accomplished using the labs ( ) layer

```
+ labs(title="Scatterplot",
       x = "Sepal width (cm)",
       y = "Sepal length (cm)")
```

lines 154 - 159

Plots in R (ggplot2)

4. **Change the *theme*:** Themes control the display of all non-data elements of the plot (font color, font size, background, gridlines, etc.)

```
+ theme ( )
```

lines 161 - 169

Plots in R (ggplot2)

5. **The facets:** add panels to display subsets of the data

```
+ facet_wrap(~Species, ncol=3)
```

lines 171 - 178

Plots in R (ggplot2)

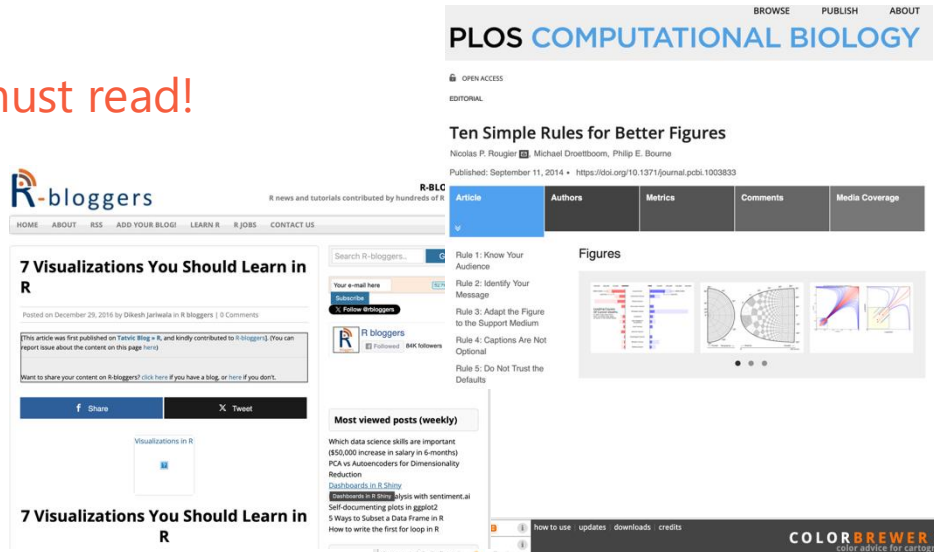
6. **And finally, save the plot**

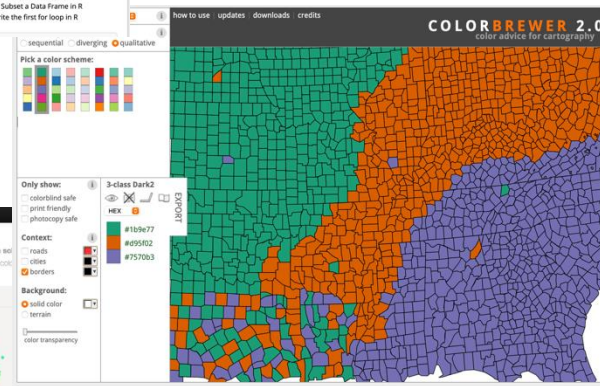
ggsave(“Myplot.pdf”)

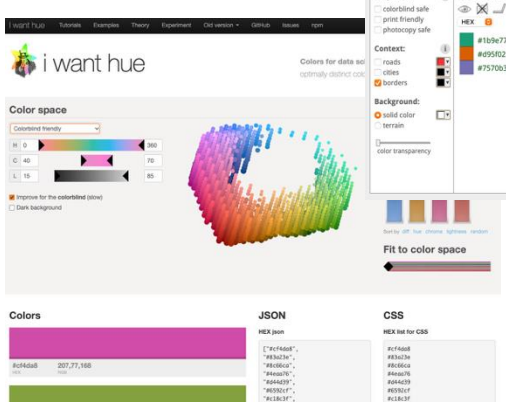
lines 180 - 181

Useful Online Resources:

- Ten simple rules for better figures - must read!
- From Data-to-viz
- R Graph Gallery
- 7 visualizations you should learn (r-bloggers)
- Selected tools for Data visualization
- Viridis color maps
- R ColorBrewer

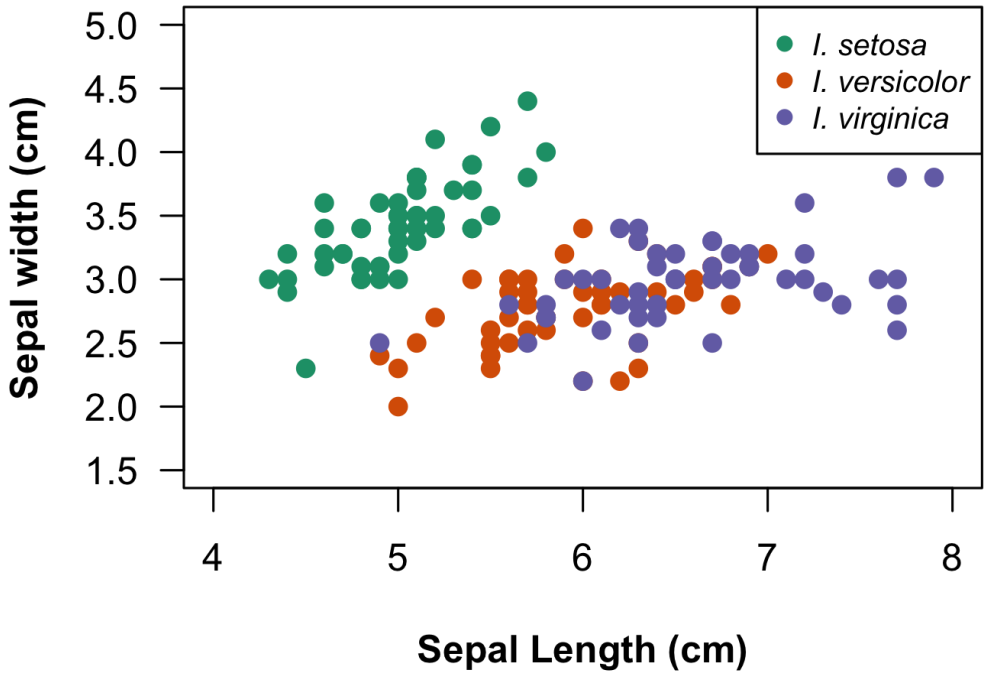




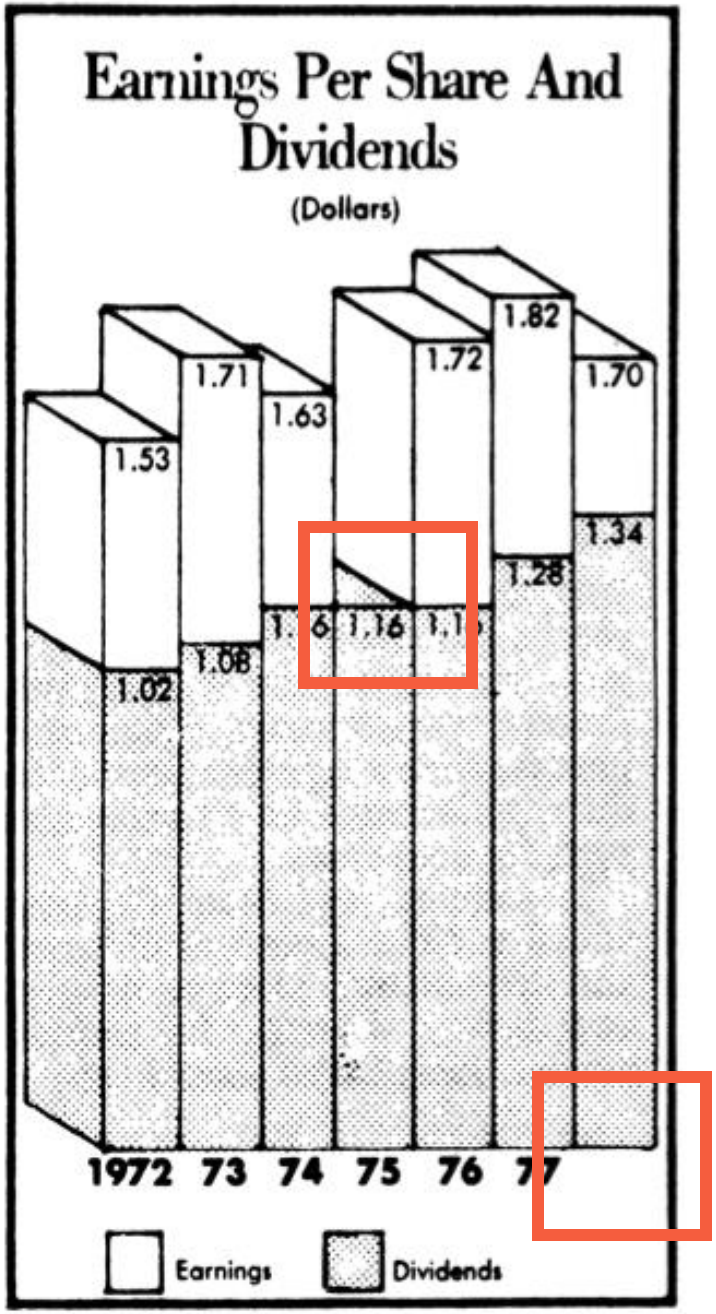


Let's make a plot

We will work on the iris dataset (again!) and make a scatterplot of sepal width (y-axis) vs. sepal length (x-axis). We will look at differences among species using points with different colors. Your plot should look like this:



Additional Examples



Less is more

Increasing the number of dimensions to a graph makes it more confusing

Even for the person who made it!

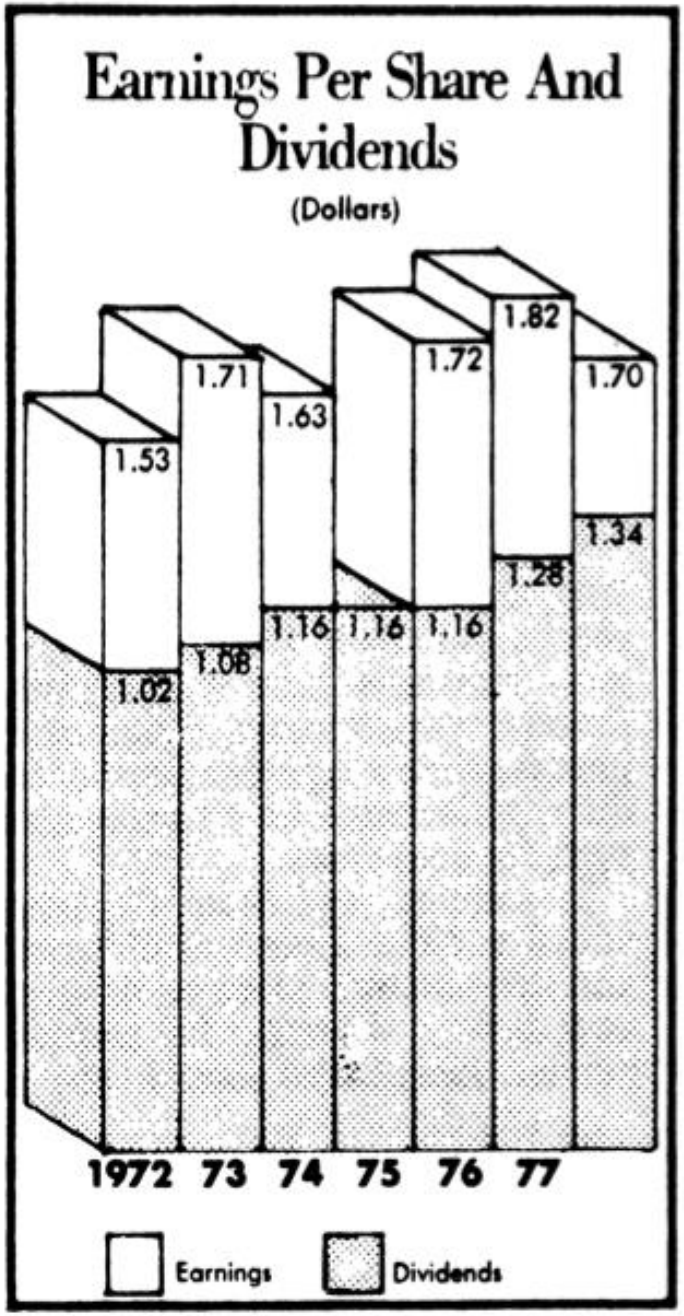


Figure 23.
(© 1979, The Washington Post).

Less is more

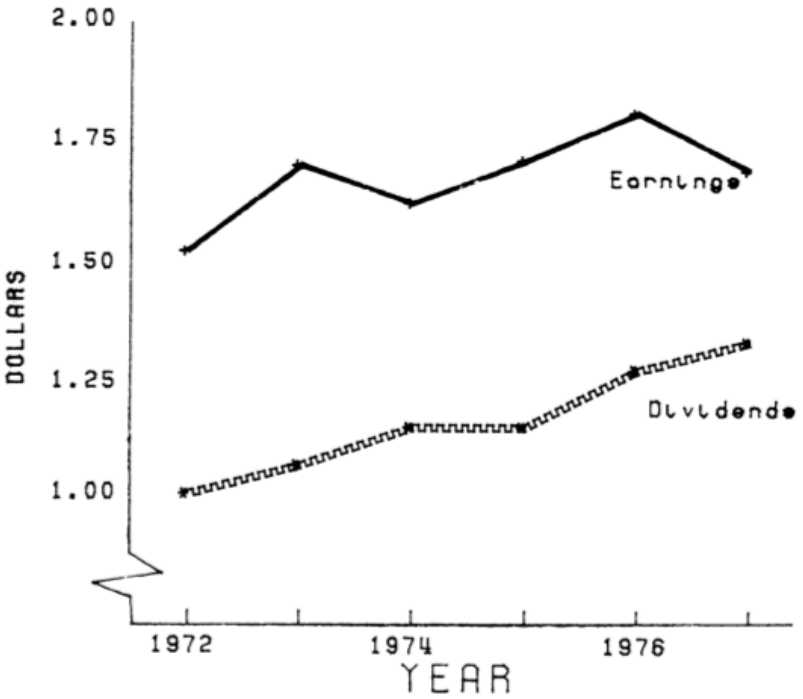
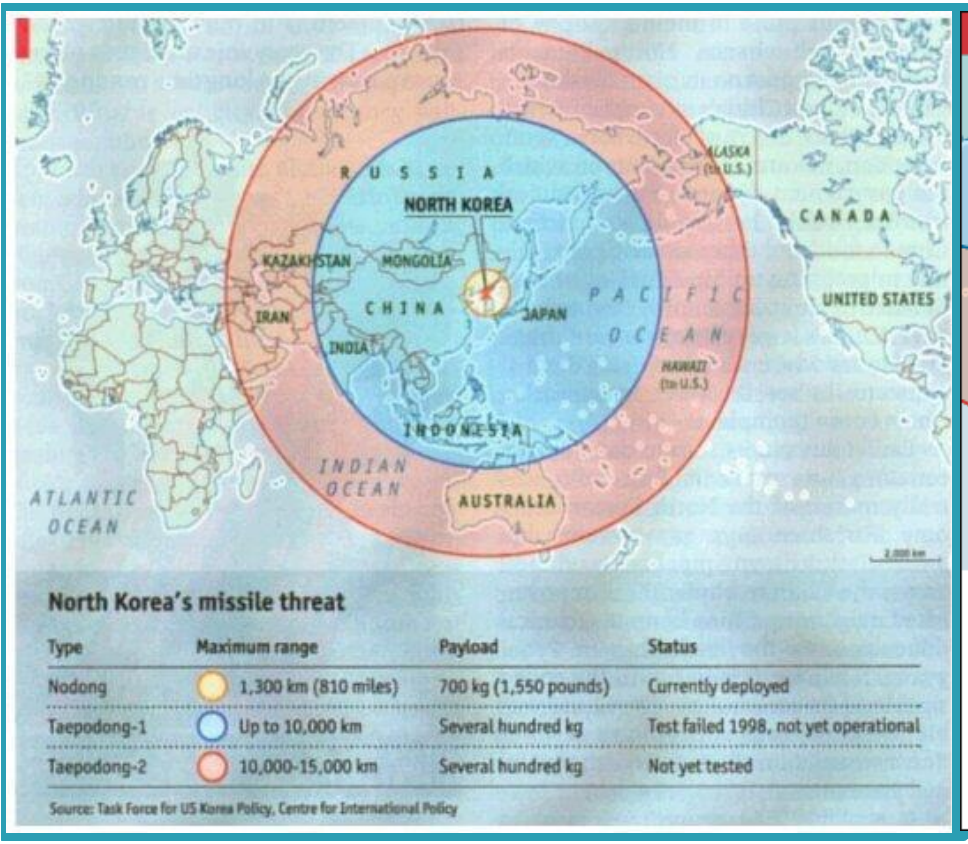
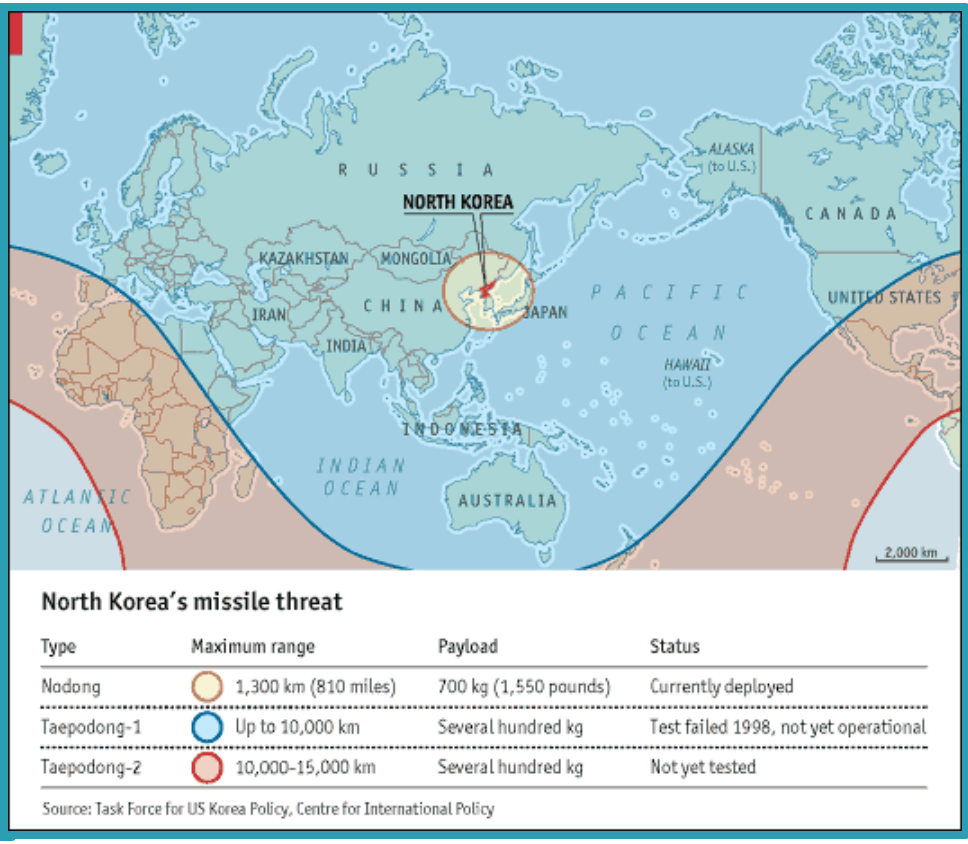


Figure 24. Data from Figure 23 redrawn simply

Precision and accuracy



The Economist

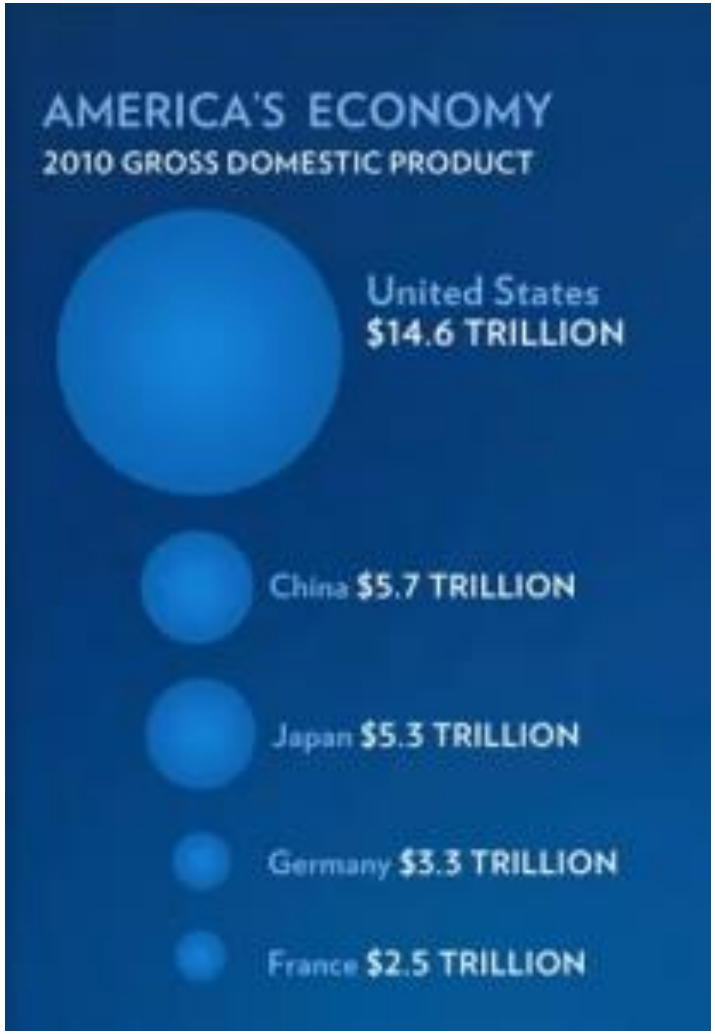


The Economist, a week later

Precision and accuracy

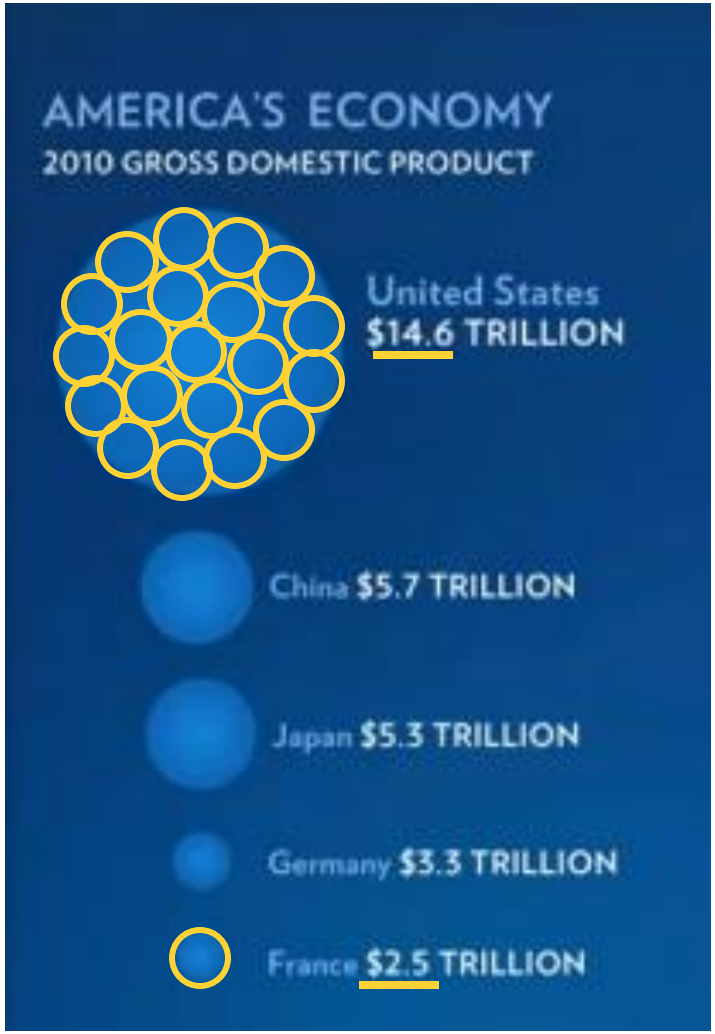
- Do not take data out of context
- Show variation in the data
- Label as necessary to make things clear
- Use appropriate (and comparable) units of measure

Precision and accuracy



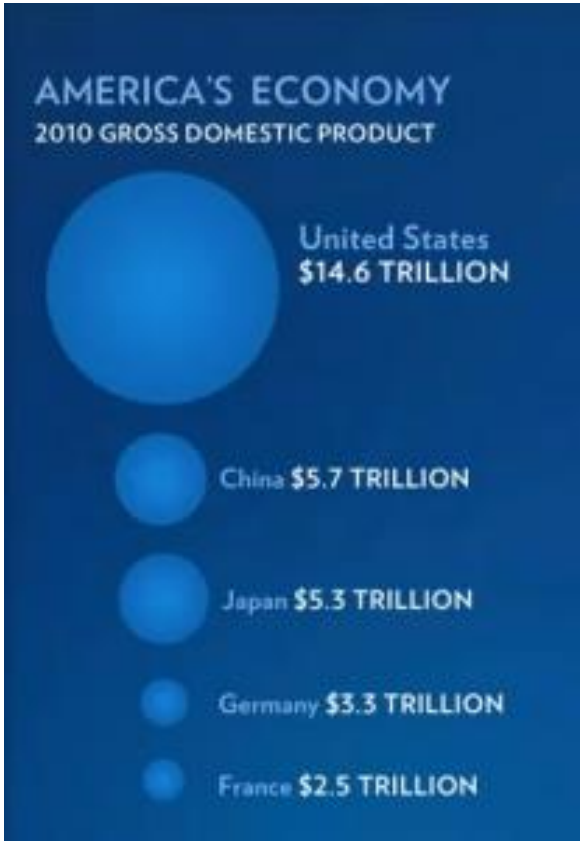
This plot may deceive viewers because the differences in areas don't correspond to the differences in the numbers presented

Precision and accuracy



This plot may deceive viewers because the differences in areas don't correspond to the differences in the numbers presented

Precision and accuracy



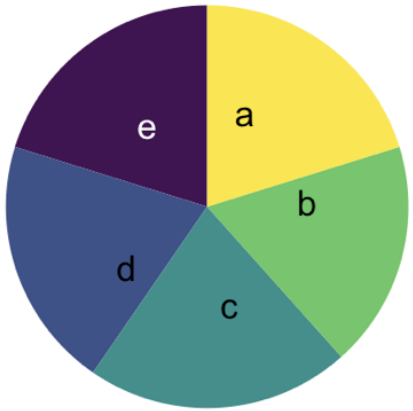


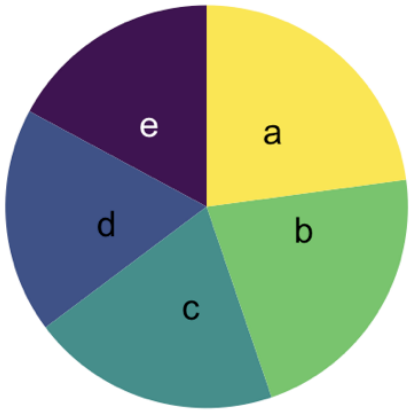
Corrected

Precision and accuracy

Avoid pie charts







Humans are bad at reading angles. On each pie, which group is the biggest one? Try to order them by value.

Precision and accuracy

